

COL216 Assignment 3 Report

Design Decisions:

The following design decisions were made during the development of the cache simulator:

1. C++ was used as the programming language because of its flexibility and performance.
2. The cache simulator implements a 2-level cache hierarchy with L1 and L2 caches.
3. The cache simulator is designed to work with an arbitrary n-way set associative cache.
4. The cache simulator reads trace files and assigns requests to the L1 cache.
5. The L1 cache sends read/write requests to the L2 cache, which interacts with DRAM.
6. Both L1 and L2 caches keep track of their own counters, such as reads, writes, misses, and hits.
7. At the end of the simulation, the program prints the statistics for both caches on the console.
8. We have incremented various parameters according to the following assumptions :
 - L2_writeback is incremented when a block from L2 is evicted with dirty bit 1.
 - Memory writes are incremented when L2_writeback takes place.
 - L1_writeback and L2_writes are incremented when dirty block from L1 is evicted .

OBSERVATIONS FROM SIMULATOR GRAPHS –

1. Varying Block Size (Fixed L1 Size, L1 Associativity, L2 Size, and L2 Associativity):
As the block size increases, the total access time generally decreases. This is because a larger block size allows for more data to be fetched in a single cache access, reducing the number of memory accesses required. Smaller block sizes (e.g., 8 or 16) result in higher total access times, as they cause more frequent cache misses and subsequent memory accesses. The effect of block size on total access time is more pronounced for memory-intensive traces that exhibit a high number of cache misses.
2. Varying L1 Size (Fixed Block Size, L1 Associativity, L2 Size, and L2 Associativity):
Increasing the L1 cache size generally leads to lower total access times. A larger cache size allows for a higher cache hit rate, reducing the number of memory accesses. The impact of L1 size on total access time is more significant for traces with a higher number of cache accesses.
3. Varying L1 Associativity (Fixed Block Size, L1 Size, L2 Size, and L2 Associativity):
Increasing the L1 associativity tends to decrease the total access time. Higher associativity improves cache utilization and reduces cache conflicts, leading to a higher cache hit rate. The effect of L1 associativity on total access time becomes more pronounced for traces with higher levels of cache contention and conflicts.
4. Varying L2 Size (Fixed Block Size, L1 Size, L1 Associativity, and L2 Associativity):

Increasing the L2 cache size generally results in lower total access times. A larger L2 cache can accommodate more data and provide a higher cache hit rate, reducing the need for accessing the main memory. The impact of L2 size on total access time is more prominent when the L1 cache is relatively smaller or when the traces exhibit a higher number of L2 cache accesses.

5. Varying L2 Associativity (Fixed Block Size, L1 Size, L1 Associativity, and L2 Size):

Increasing the L2 associativity tends to decrease the total access time. Higher associativity reduces cache conflicts in the L2 cache and improves its hit rate, resulting in fewer memory accesses. The effect of L2 associativity on total access time becomes more significant when the traces have higher levels of contention for the L2 cache.

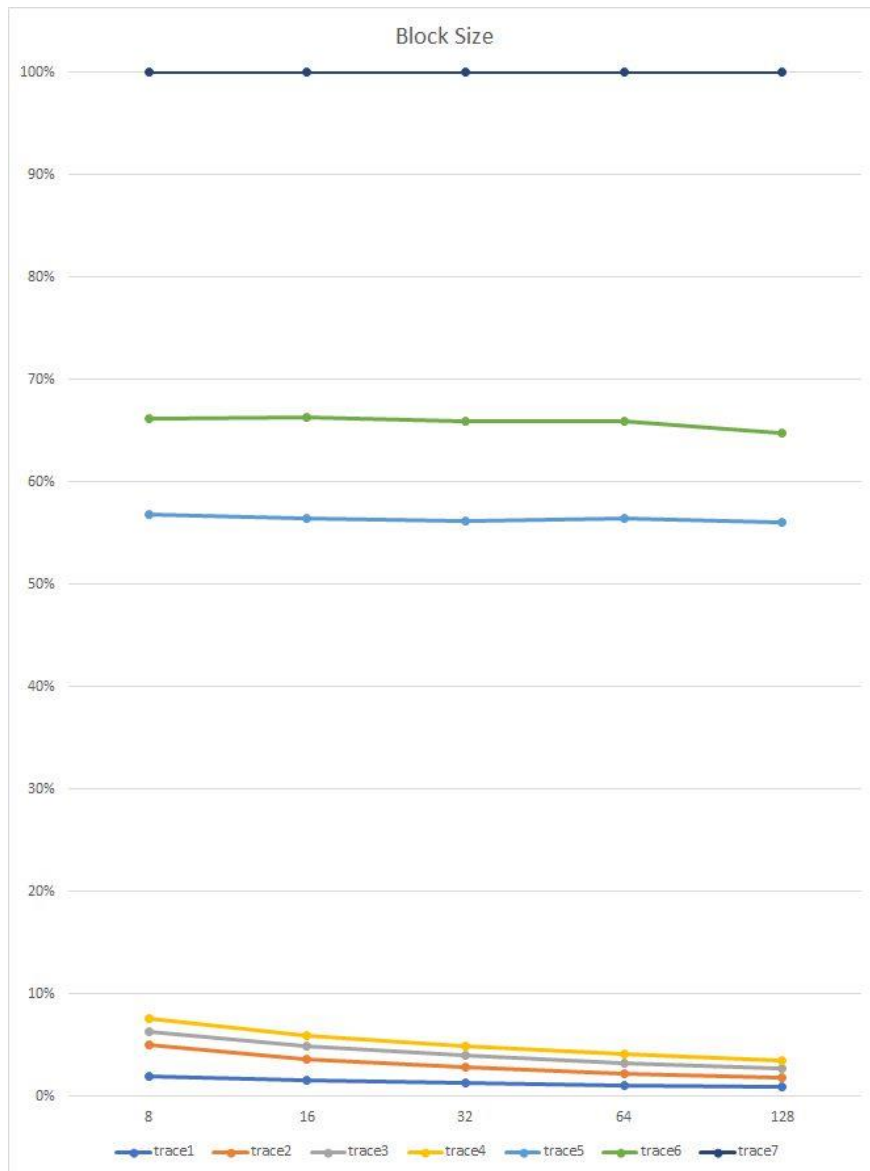
GENERAL OBSERVATIONS-

The total access time is directly influenced by cache performance, including cache hit rate and the level of cache hierarchy accessed. Smaller caches or caches with lower associativity tend to exhibit higher total access times due to increased cache misses and subsequent memory accesses. Traces with a higher number of cache accesses or cache contention tend to have longer total access times. Increasing cache sizes, associativity, or block sizes generally leads to lower total access times by reducing cache misses and memory accesses. The impact of cache parameters on total access time can vary depending on the characteristics of the trace files, such as their access patterns and memory locality.

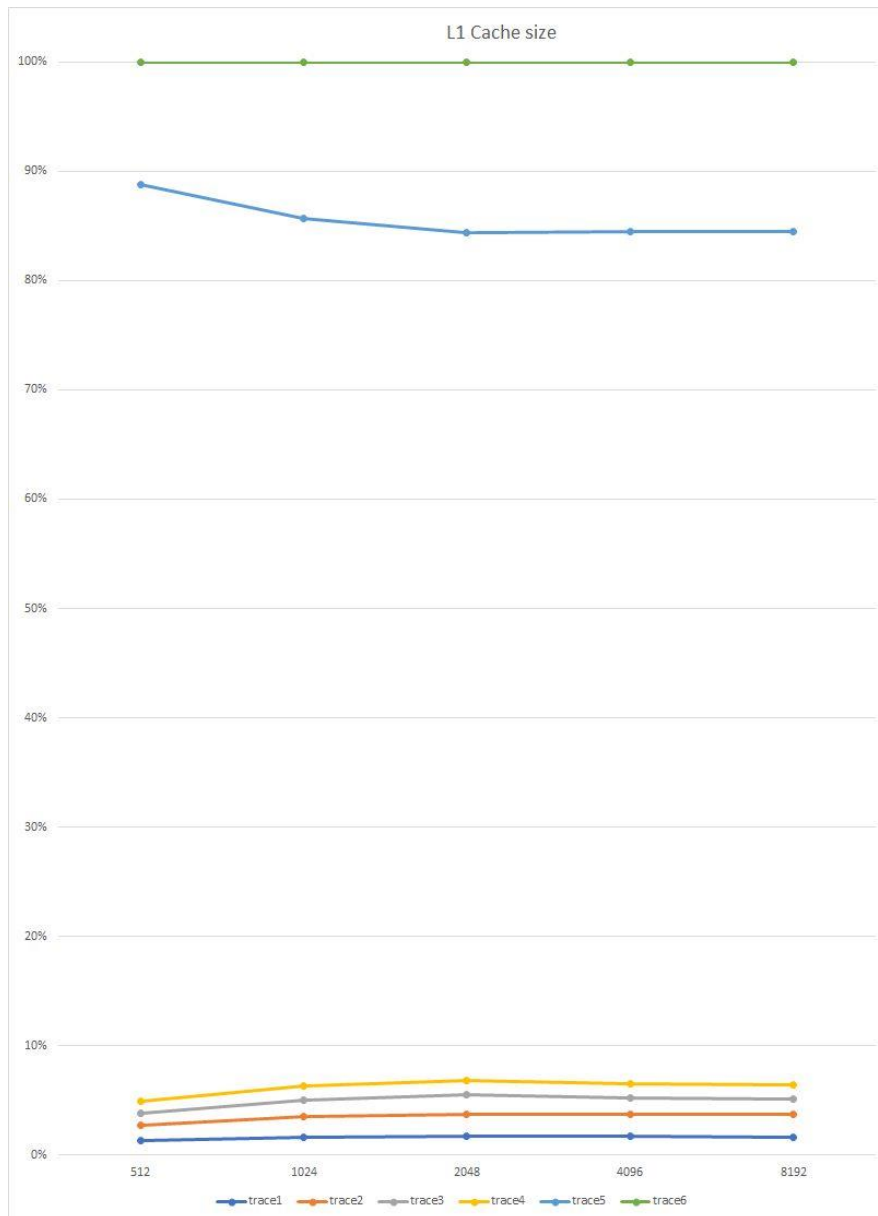
Note that for each of the following graphs the y axis contains time in milliseconds.

GRAPHS –

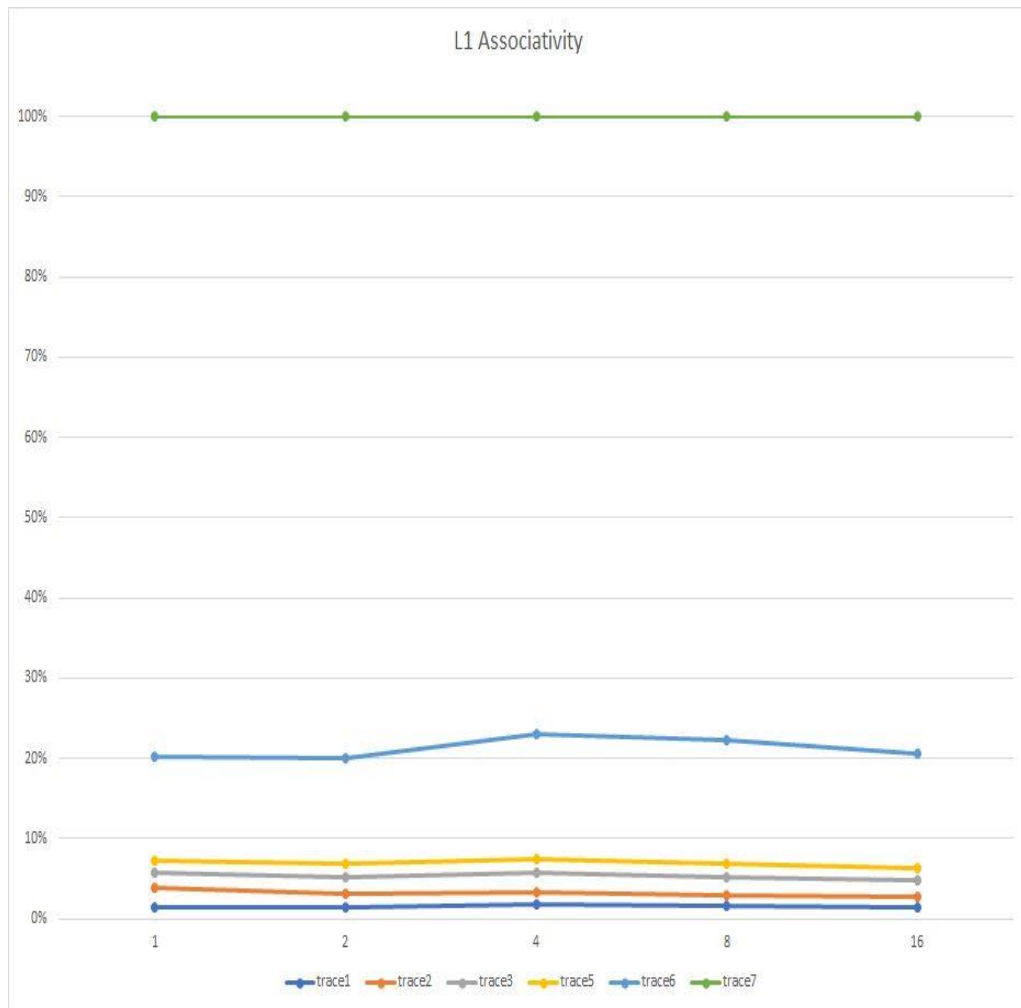
1. Time Vs BlockSize :



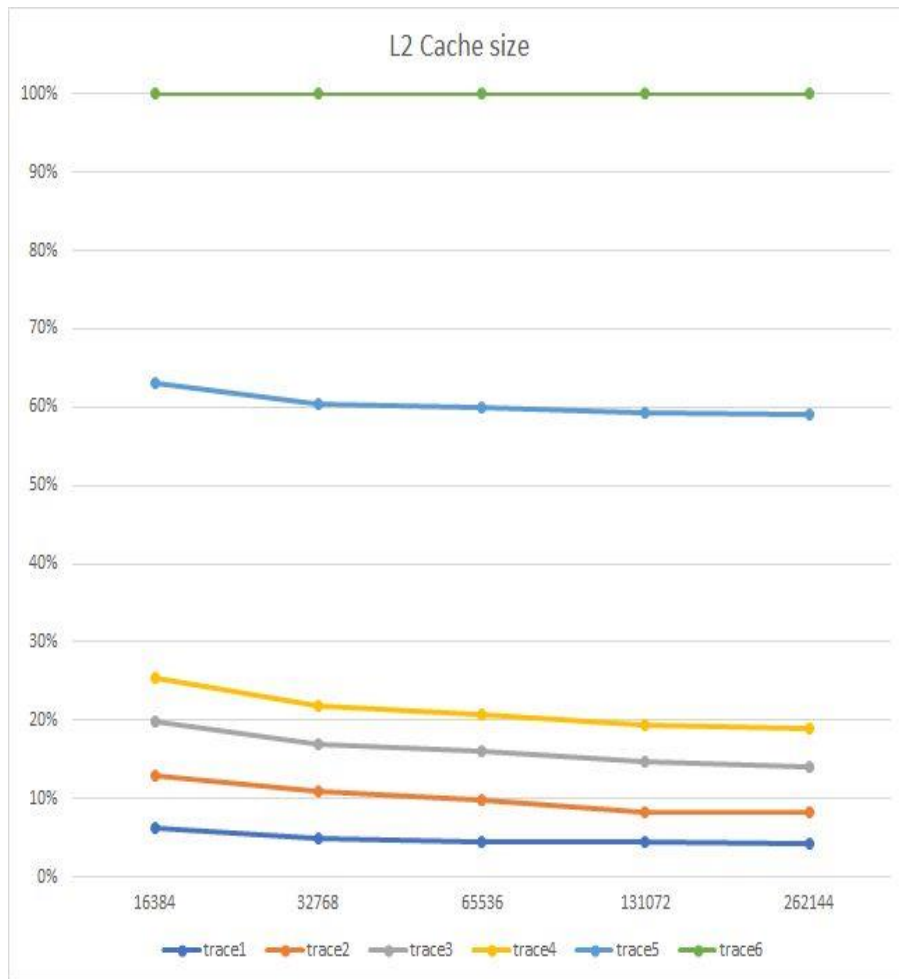
2. Time Vs L1 Cache Size:



3. Time Vs L1 Associativity:



4. Time Vs L2 Cache Size :



5. Time Vs L2 Associativity:

